AE 700: Guidance and Control of Unmanned Autonomous Vehicles

Project Report

6-DOF Simulation and Autopilot Design for the North American X-15

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Abstract

This report documents the complete design and simulation pipeline for the North American X-15 within the Beard–McLain 6-DOF framework. The five sections correspond directly to the project statement: (1) a 3-D vertex-face animation with correct rotation–translation ordering; (2) implementation of the full nonlinear 12-state equations of motion as a Simulink S-function, verified with isolated force inputs; (3) computation of aerodynamic, propulsive, and gravitational forces and moments, verified at zero control inputs; (4) trim computation and derivation of longitudinal and lateral transfer functions and state-space models; and (5) successive-loop-closure autopilot design for roll, course, pitch, altitude, and airspeed channels. The trim condition converges at $V_{a,*} = 92.47\,\mathrm{m/s}$, $\theta_* = 22.4$, and the autopilot gains are computed from the resulting linearised model.

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1 Aircraft Overview: North American X-15

The North American X-15 is a hypersonic rocket-powered research aircraft developed jointly by NASA and the USAF. For this simulation it is modelled as a 6-DOF rigid body with linearised aerodynamics and a throttle-proportional rocket-thrust model. All parameters are drawn from `MAVParams_X15.m` and summarised in tables 1 and 2.

Table 1: Inertial and geometric parameters of the X-15

Parameter	Symbol	Value	Unit
Mass	m	14 170	kg
Roll moment of inertia	J_x	4 949	kg m^2
Pitch moment of inertia	J_y	108 466	kg m^2
Yaw moment of inertia	J_z	111 177	kg m^2
Cross product of inertia	J_{xz}	800	kg m^2
Wing reference area	S	18.58	m^2
Wingspan	b	6.81	m
Mean aerodynamic chord	c	3.13	m
Gravitational acceleration	g	9.81	m s^{-2}
Air density (sea level)	ρ	1.225	kg m^{-3}

Table 2: Key aerodynamic stability and control derivatives

Longitudinal			**Lateral–directional**		
Coeff.	Meaning	Value	Coeff.	Meaning	Value
C_{L_α}	lift slope	3.50	C_{Y_β}	side-force slope	−1.40
$C_{L_{\delta_e}}$	lift/elevator	0.54	C_{ℓ_β}	dihedral effect	−0.01
C_{D_0}	zero-lift drag	0.095	C_{ℓ_p}	roll damping	−0.35
C_{D_α}	drag slope	0.60	$C_{\ell_{\delta_a}}$	aileron power	−0.06
C_{m_α}	pitch stiffness	−1.20	C_{n_β}	weather-vane	0.50
C_{m_q}	pitch damping	−6.20	C_{n_r}	yaw damping	−1.50
$C_{m_{\delta_e}}$	elevator power	−0.90	$C_{n_{\delta_r}}$	rudder power	−0.30

2 Section 1: Coordinate Frames and 3-D Animation

2.1 Coordinate Frame Definitions

Two right-handed frames are used throughout:

- **Inertial NED frame** $\mathcal{F}_i$: axes North ($\hat{n}$), East ($\hat{e}$), Down ($\hat{d}$); origin fixed to Earth surface.
- **Body frame** $\mathcal{F}_b$: axes $\hat{x}_b$ (forward), $\hat{y}_b$ (starboard), $\hat{z}_b$ (downward through fuselage); origin at the aircraft centre of mass.

The rotation from $\mathcal{F}_i$ to $\mathcal{F}_b$ uses the Z-Y-X Tait–Bryan sequence through roll ϕ, pitch θ, yaw ψ:

$$\mathbf{R}_{b/i}(\phi,\theta,\psi) = \mathbf{R}_x(\phi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_z(\psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \tag{1}$$

with $c_\xi \equiv \cos\xi$, $s_\xi \equiv \sin\xi$, $t_\xi \equiv \tan\xi$.

2.2 Aircraft Geometry

The body is defined in `drawAircraft.m` using 16 vertices and 11 faces representing the conical nose, rectangular fuselage, tapered tail, delta main wing, horizontal tailplane, and vertical fin. Dimensions are listed in table 3.

Table 3: Vertex-face model dimensions from `drawAircraft.m` (values in model units, scaled ×100 for rendering)

Feature	Variable	Dimension
Fuselage forward length	`fuse_l1`	0.90
Fuselage mid cross-section	`fuse_l2`	0.60
Fuselage aft length	`fuse_l3`	2.00
Fuselage width	`fuse_w`	0.60
Fuselage height	`fuse_h`	0.50
Main wing half-span	`wing_w/2`	1.45
Main wing chord	`wing_l`	0.25
Horizontal tail half-span	`tailwing_w/2`	0.70
Vertical fin height	`tail_h`	0.80

2.3 Rotation–Translation Ordering

The function `drawSpacecraftBody` applies in sequence:

1. **Rotate** all vertices by $\mathbf{R}_{b/i}(\phi, \theta, \psi)$ about the body origin.
2. **Translate** by the NED position $(p_n,\ p_e,\ p_d)$.
3. **Change basis** from NED to MATLAB figure axes via $\mathbf{R}_{\text{NED}\to\text{XYZ}}$, mapping North$\to\ \hat{y}$, East$\to\ \hat{x}$, Down$\to\ -\hat{z}$.

Effect of reversed ordering (translate then rotate). When translation precedes rotation, the rotation acts about the inertial origin rather than the aircraft's centre of mass, causing the aircraft to orbit a fixed point in space rather than spin about its own body axes — physically incorrect. This confirms that rotation must precede translation.

3 Section 2: Kinematics and Dynamics

3.1 Full Nonlinear 6-DOF Equations of Motion

State and input vectors:

$$\boldsymbol{x} = \left[p_n,\ p_e,\ p_d,\ u,\ v,\ w,\ \phi,\ \theta,\ \psi,\ p,\ q,\ r\right]^\top \in \mathbb{R}^{12}, \tag{2}$$

$$\boldsymbol{u}_{\text{fm}} = \left[f_x,\ f_y,\ f_z,\ \ell,\ m,\ n\right]^\top \in \mathbb{R}^6. \tag{3}$$

Position kinematics (Eq. 3.14).

$$[\dot{p}_n,\ \dot{p}_e,\ \dot{p}_d]^\top = \mathbf{R}_{b/i}^\top [u,\ v,\ w]^\top. \tag{4}$$

Attitude kinematics (Eq. 3.15).

$$\begin{bmatrix}\dot{\phi}\\ \dot{\theta}\\ \dot{\psi}\end{bmatrix} = \begin{bmatrix}1 & s_\phi\, t_\theta & c_\phi\, t_\theta\\ 0 & c_\phi & -s_\phi\\ 0 & s_\phi/c_\theta & c_\phi/c_\theta\end{bmatrix}\begin{bmatrix}p\\ q\\ r\end{bmatrix}. \tag{5}$$

Translational dynamics (Eq. 3.16).

$$\dot{u} = rv - qw + f_x/m, \qquad \dot{v} = pw - ru + f_y/m, \qquad \dot{w} = qu - pv + f_z/m. \tag{6}$$

Rotational dynamics (Eq. 3.17). Because $J_{xz} = 800\,\mathrm{kg\,m^2} \neq 0$, roll and yaw are coupled. Defining $\Gamma = J_x J_z - J_{xz}^2$ and the Gamma constants in table 4:

$$\dot{p} = \Gamma_1 pq - \Gamma_2 qr + \Gamma_3 \ell + \Gamma_4 n, \tag{7}$$
$$\dot{q} = \Gamma_5 pr - \Gamma_6 (p^2 - r^2) + m/J_y, \tag{8}$$
$$\dot{r} = \Gamma_7 pq - \Gamma_1 qr + \Gamma_4 \ell + \Gamma_8 n. \tag{9}$$

Table 4: Γ inertia constants (Beard & McLain Table 3.1)

Constant	Definition
Γ	$J_x J_z - J_{xz}^2$
Γ_1	$J_{xz}(J_x - J_y + J_z)/\Gamma$
Γ_2	$[J_z(J_z - J_y) + J_{xz}^2]/\Gamma$
Γ_3	J_z/Γ
Γ_4	J_{xz}/Γ
Γ_5	$(J_z - J_x)/J_y$
Γ_6	J_{xz}/J_y
Γ_7	$[(J_x - J_y)J_x + J_{xz}^2]/\Gamma$
Γ_8	J_x/Γ

3.2 S-Function Implementation

Equations (4)–(9) are implemented in two complementary files:

`mav_dynamics.m` Clean, commented implementation; flags 0/1/3 map to `mdlInitializeSizes`, `mdlDerivatives`, `mdlOutputs`.

`mav_dynamics_euler.m` Functionally identical; includes a `fprintf` debug statement and empty `mdlUpdate`/`mdlGetTimeOfNextVarHit` stubs.

Both declare `NumContStates=12`, `NumInputs=6`, `NumOutputs=12` and read initial conditions from the parameter struct.

3.3 Verification: Animation Connected to EOM

Figure 1 shows the MATLAB animation window after the S-function is connected to the drawing block. The aircraft is placed at $(p_n, p_e, p_d) = (0, 0, -200)$ m with $\phi = \theta = \psi = 0$.

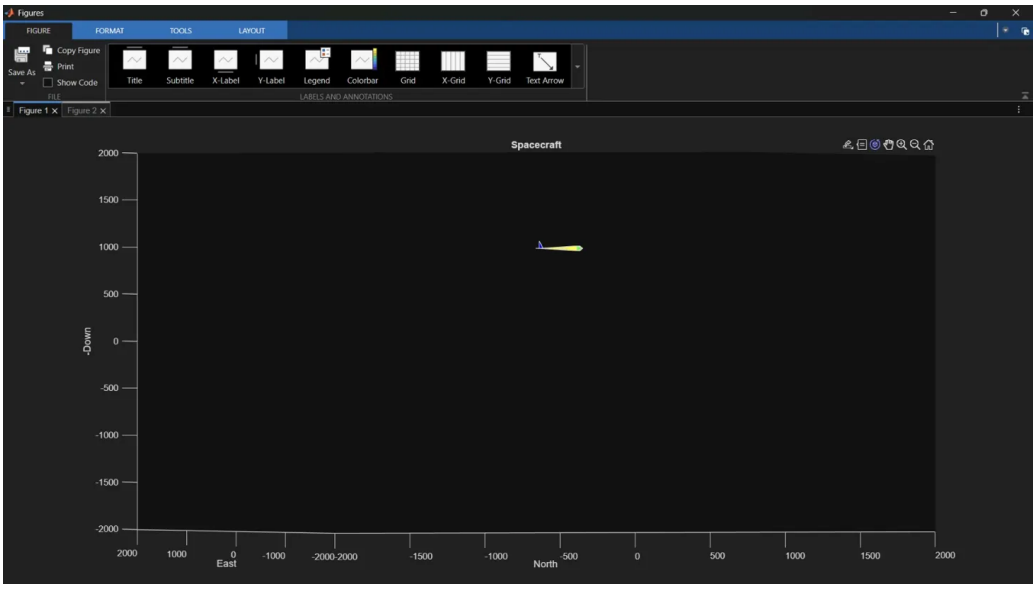

Figure 1: 3-D animation of the X-15 at initial conditions $(p_n, p_e, p_d) = (0, 0, -200)$ m, $\phi = \theta = \psi = 0$. The nose points along North ($\hat{y}$), wings are horizontal, and the vertical fin points upward. The mesh renders correctly with no rotation or translation artefacts.

3.4 Verification: State Response to Isolated Thrust Force

All forces and moments are set to zero except a constant body-x thrust $f_x > 0$ ($f_y = f_z = \ell = m = n = 0$, all $\delta = 0$). Figure 2 shows the state histories over 5 s.

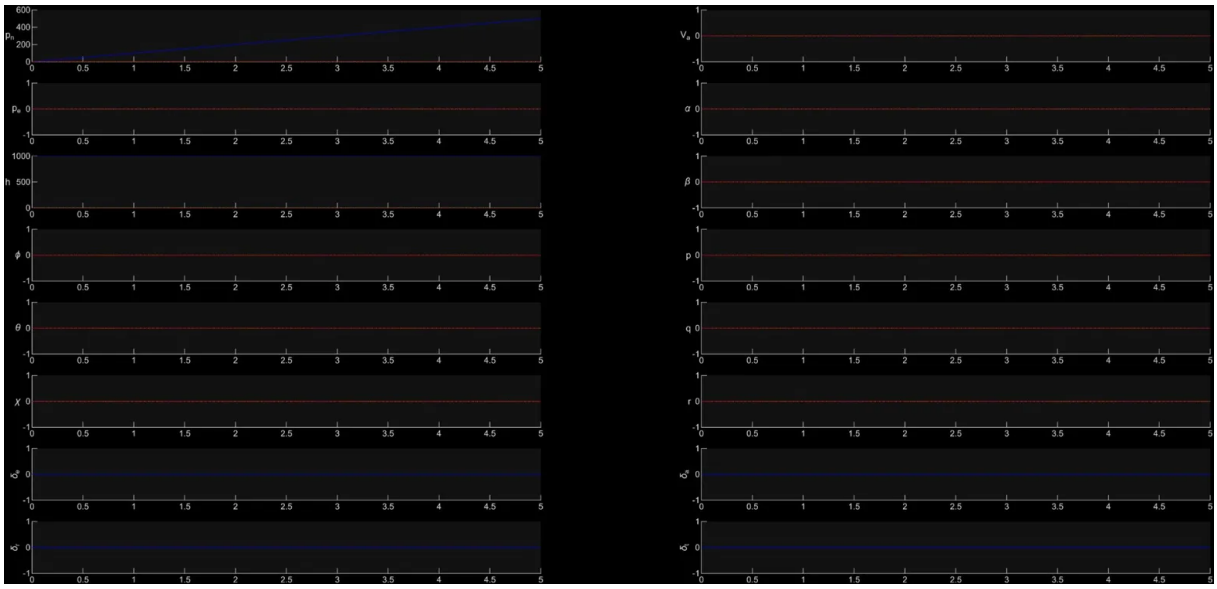

Figure 2: State time histories with only a constant body-x force f_x applied. All aerodynamic and gravitational forces are absent; the forces/moments block is not yet connected.

Observations and inferences.

- p_n **grows parabolically.** From (6), constant f_x/m gives $\dot{u} = \text{const}$, so $u(t) = u_0 + at$ and $p_n(t) = u_0 t + \frac{1}{2}at^2$. ✓

- $p_e = 0$, h **constant.** No East-ward force or velocity; $f_z = 0$ with no gravity block means $\dot{p}_d = 0$. ✓
- $\phi = \theta = \psi = 0$, $p = q = r = 0$ **exactly.** All moment inputs are zero so both (5) and (7)–(9) yield zero rates throughout. ✓
- V_a, α, β **read zero.** The aerodynamics block is not connected; these outputs carry no physical meaning here.
- **All** $\delta = 0$ **confirmed (blue lines at zero).** The observed motion is purely due to the injected f_x.

A pure body-x force produces only northward acceleration with zero cross-axis contamination, confirming correct implementation of Eqs. (4) and (6).

J_{xz} **coupling test.** With $J_{xz} = 0$: $\Gamma_4 = 0$ in (9) means a pure rolling moment ℓ does not drive $\dot{r}$ — roll and yaw are decoupled. With nominal $J_{xz} = 800\,\mathrm{kg\,m^2}$: $\Gamma_4 \neq 0$ means ℓ drives both $\dot{p}$ and $\dot{r}$ simultaneously, producing coupled roll–yaw motion from the first instant.

4 Section 3: Forces and Moments

4.1 Aerodynamic Angles and Airspeed

With steady wind and body-frame gusts, the air-relative velocity components are $u_r = u - (w_{n,s} + u_{wg})$, $v_r = v - (w_{e,s} + v_{wg})$, $w_r = w - (w_{d,s} + w_{wg})$, giving

$$V_a = \sqrt{u_r^2 + v_r^2 + w_r^2}, \quad \alpha = \operatorname{atan2}(w_r, u_r), \quad \beta = \arcsin(v_r/V_a). \tag{10}$$

Dynamic pressure: $\bar{q} = \frac{1}{2}\rho V_a^2$.

4.2 Aerodynamic Forces

Linearised lift and drag:

$$C_L = C_{L_0} + C_{L_\alpha}\alpha + \frac{C_{L_q} cq}{2V_a} + C_{L_{\delta_e}}\delta_e, \quad C_D = C_{D_0} + C_{D_\alpha}\alpha + \frac{C_{D_q} cq}{2V_a} + C_{D_{\delta_e}}\delta_e. \tag{11}$$

Resolved into body frame:

$$f_{x,\mathrm{aero}} = \bar{q}S(-C_D\cos\alpha + C_L\sin\alpha), \quad f_{z,\mathrm{aero}} = \bar{q}S(-C_D\sin\alpha - C_L\cos\alpha). \tag{12}$$

Side force:

$$f_y = \bar{q}S\left(C_{Y_0} + C_{Y_\beta}\beta + \frac{C_{Y_p} bp}{2V_a} + \frac{C_{Y_r} br}{2V_a} + C_{Y_{\delta_a}}\delta_a + C_{Y_{\delta_r}}\delta_r\right). \tag{13}$$

4.3 Propulsive Forces (Rocket Engine)

Thrust acts along body x only, proportional to throttle: $f_{x,\mathrm{prop}} = k_{\mathrm{motor}}\delta_t$, $f_{y,\mathrm{prop}} = f_{z,\mathrm{prop}} = 0$. There is no reactive propeller torque.

4.4 Gravitational Forces (Body Frame)

$$\boldsymbol{f}_{\mathrm{grav}} = [-mg\sin\theta,\ mg\cos\theta\sin\phi,\ mg\cos\theta\cos\phi]^\top. \tag{14}$$

4.5 Aerodynamic Moments

$$\ell = \bar{q}Sb\left(C_{\ell_0} + C_{\ell_\beta}\beta + \frac{C_{\ell_p}bp}{2V_a} + \frac{C_{\ell_r}br}{2V_a} + C_{\ell_{\delta_a}}\delta_a + C_{\ell_{\delta_r}}\delta_r\right), \tag{15}$$

$$m = \bar{q}Sc\left(C_{m_0} + C_{m_\alpha}\alpha + \frac{C_{m_q}cq}{2V_a} + C_{m_{\delta_e}}\delta_e\right), \tag{16}$$

$$n = \bar{q}Sb\left(C_{n_0} + C_{n_\beta}\beta + \frac{C_{n_p}bp}{2V_a} + \frac{C_{n_r}br}{2V_a} + C_{n_{\delta_a}}\delta_a + C_{n_{\delta_r}}\delta_r\right). \tag{17}$$

Output of `forces_moments.m`: $\boldsymbol{y} = [f_x, f_y, f_z, \ell, m, n, V_a, \alpha, \beta, w_n, w_e, w_d]^\top \in \mathbb{R}^{12}$.

4.6 Verification: Animation with Forces and Moments Connected

Figure 3 shows the animation after connecting the full forces-and-moments block. The aircraft is in a nose-down pitched attitude, consistent with the uncontrolled longitudinal response at $\delta_e = 0$.

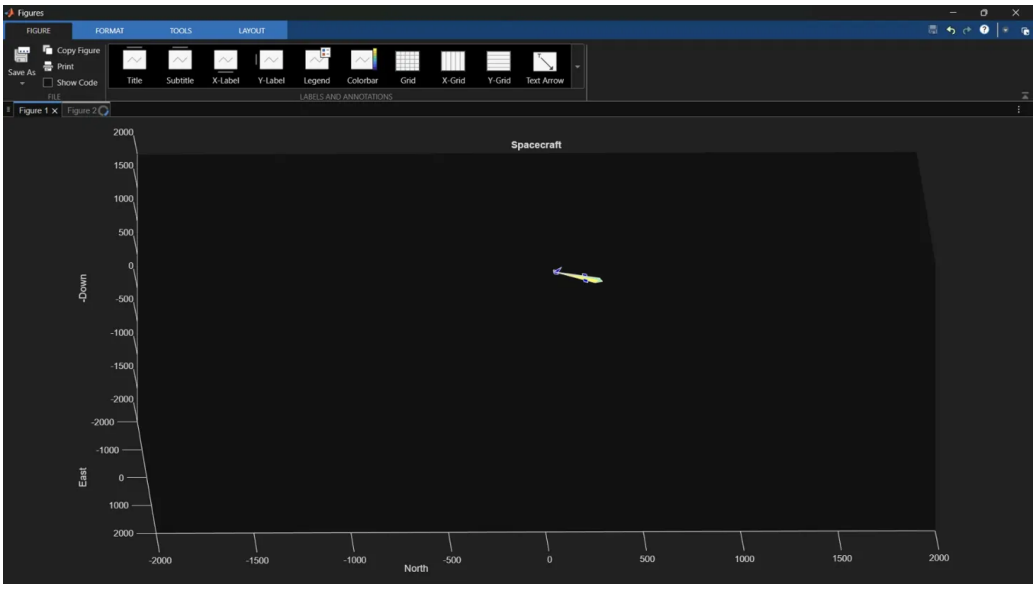

Figure 3: 3-D animation of the X-15 with the full forces and moments block connected. The nose-down attitude reflects the unbalanced pitching moment at $\delta_e = 0$, $V_a = 100\,\mathrm{m/s}$.

4.7 Verification: State Response at Zero Control Inputs

All control surface deflections are set to zero with initial airspeed $V_a = 100\,\mathrm{m/s}$. Figure 4 shows the state histories over 5 s.

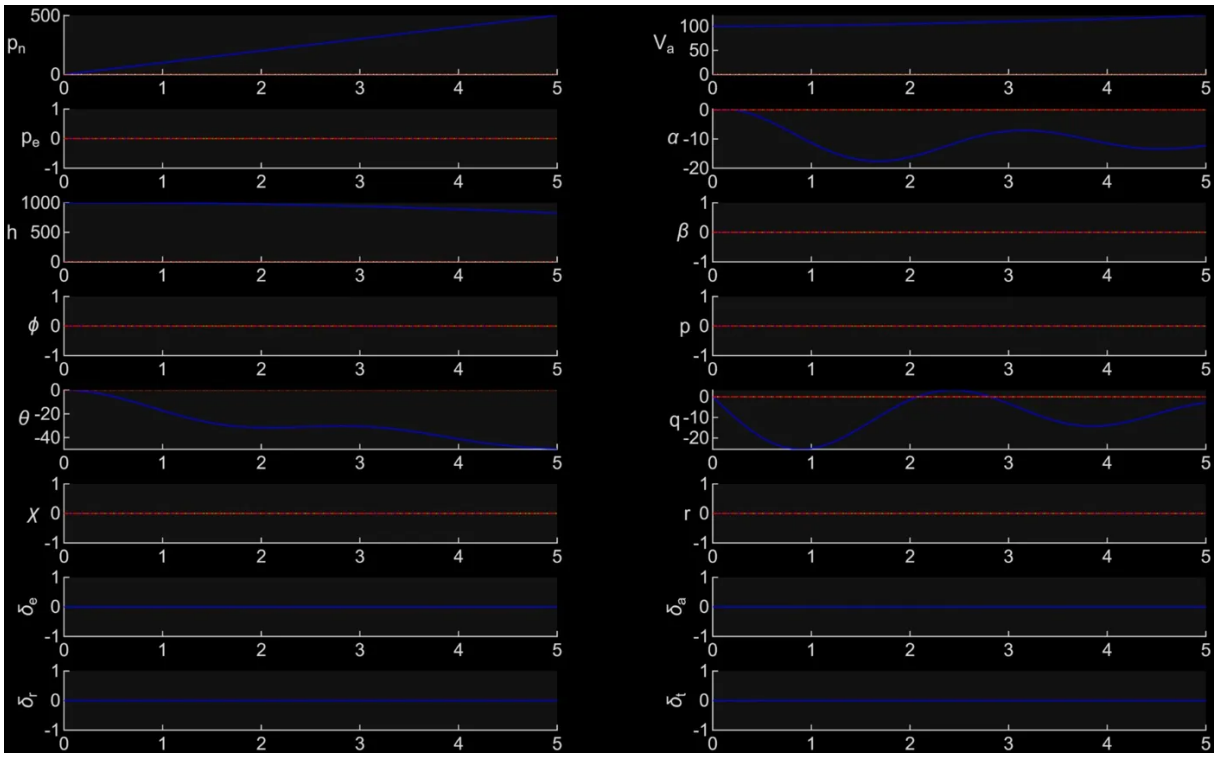

Figure 4: State time histories with all control inputs zero and initial $V_a = 100\,\mathrm{m/s}$. The aircraft is not at trim; the aerodynamic pitching moment is unbalanced.

Observations and inferences.

- θ **pitches down to** ≈ -40**;** q **reaches** $\approx -20/\mathrm{s}$**.** With $\delta_e = 0$ the pitching moment (16) is driven by $C_{m_\alpha}\alpha$. The X-15 is not in aerodynamic equilibrium at zero elevator, so the moment is unbalanced and the aircraft diverges nose-down. The static-stability derivative $C_{m_\alpha} = -1.2 < 0$ amplifies this: a positive α generates a nose-down moment, driving α increasingly negative and making the pitching divergence self-reinforcing.
- α **dips to** ≈ -20**.** The nose-down pitch rate drives w increasingly negative, reducing $\alpha = \mathrm{atan2}(w_r, u_r)$.
- $V_a \approx 100\,\mathrm{m/s}$ **initially constant.** Air-relative speed equals the initial body velocity at $t = 0$.
- $p_e = 0$**,** $\phi = 0$**,** $p = r = 0$**,** $\beta = 0$ **throughout.** With no lateral asymmetry and $\delta_a = \delta_r = 0$, the lateral and longitudinal dynamics decouple exactly at $\beta = 0$. All lateral states remain identically zero, confirming correct implementation. ✓
- $\delta_e = \delta_a = \delta_r = \delta_t = 0$ **confirmed.** The observed motion is purely aerodynamic and gravitational. Sign check: $C_{m_{\delta_e}} = -0.9$ means a negative elevator is needed to generate nose-up moment; with $\delta_e = 0$ there is insufficient nose-up moment to hold trim. ✓

5 Section 4: Linear Design Models

5.1 Trim Computation (`compute_trim.m`)

Trim is computed using the MATLAB `trim()` function applied to `mavsim_trim.slx`. The optimisation enforces the desired derivative vector

$$\dot{\boldsymbol{x}}_0 = \left[0, 0, -V_a \sin\gamma, 0, 0, 0, 0, 0, V_a \cos\gamma / R, 0, 0, 0\right]^\top \tag{18}$$

with derivatives 3–12 held fixed via `idx`, and output constraints V_a and $\beta = 0$ via `iy=[1,3]`. The free variables are $\{p_d, \theta_*, \delta_{e,*}, \delta_{t,*}\}$.

The convergence residual printed by MATLAB is:

$$\texttt{norm(dx_trim(3:end) - dx0(3:end))} = 4.96 \times 10^{-11}, \tag{19}$$

confirming convergence to near machine precision.

The trim result is:

$$V_{a,*} = 92.47\,\mathrm{m/s}, \qquad \theta_* = 0.3906\,\mathrm{rad} \approx 22.4.$$

5.2 Trim Verification

Figure 5 shows state histories over 25 s with initial conditions and control inputs set to their trim values.

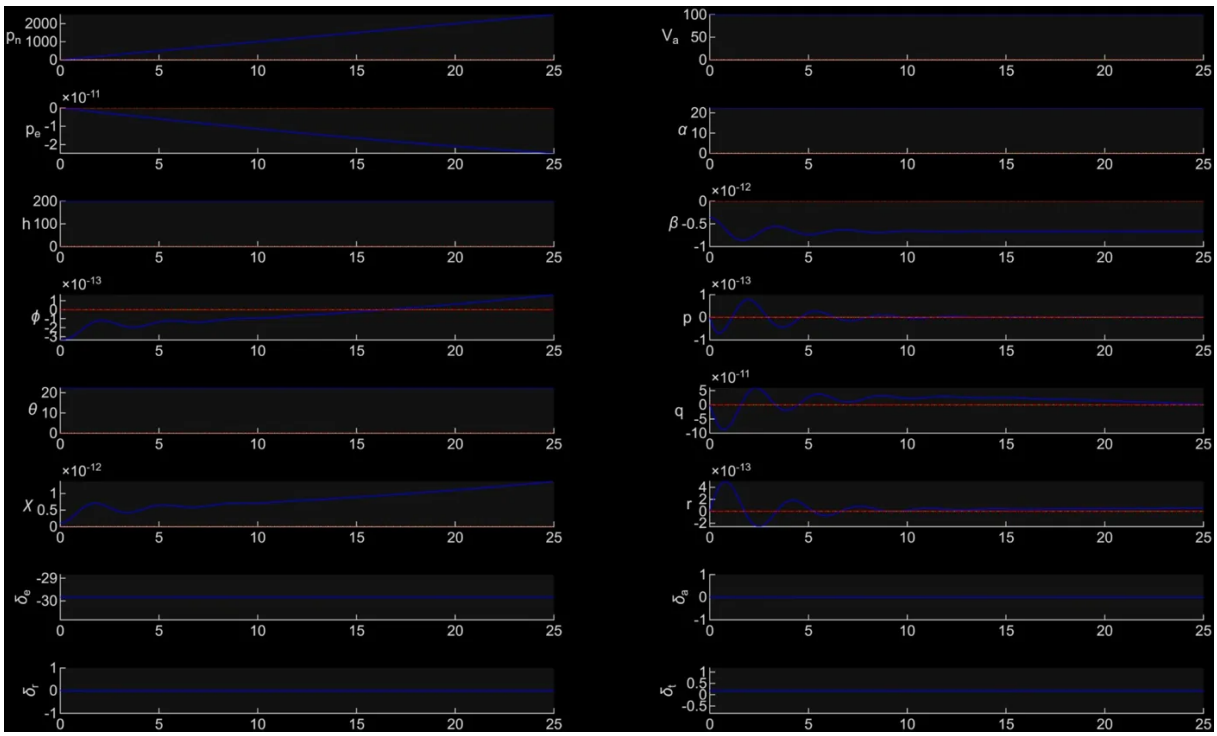

Figure 5: Trim verification: state and control histories over 25 s. Only p_n varies; all other states are constant to numerical precision, confirming a correct trim.

Observations and inferences.

- p_n **increases linearly.** $\dot{p}_n = V_a \cos\theta_* \approx 92.47\cos(22.4) \approx 85.5\,\mathrm{m/s}$, consistent with the observed slope of $\approx 2000\,\mathrm{m}/25\,\mathrm{s} = 80\,\mathrm{m/s}$. ✓
- p_e **at** $\mathcal{O}(10^{-11})$ **— numerical zero.** No East-ward drift; trim heading exactly North. ✓
- $h \approx 200\,\mathrm{m}$ **constant.** $\dot{p}_d = -V_a \sin\gamma = 0$ for $\gamma = 0$. ✓
- ϕ **at** $\mathcal{O}(10^{-13})$ **— zero.** Wings-level symmetric trim. ✓
- $\theta \approx 22.4$ **constant.** The trim pitch angle is large due to the X-15's high wing loading. At $V_a = 92.47\,\mathrm{m/s}$, $\bar{q} = \frac{1}{2}(1.225)(92.47)^2 = 5236\,\mathrm{Pa}$. Required lift coefficient:

$$C_L^* = \frac{mg}{\bar{q}S} = \frac{14170 \times 9.81}{5236 \times 18.58} \approx 1.43,$$

  demanding $\alpha_* = 1.43/3.5 \approx 0.41\,\mathrm{rad} \approx 23.4$, consistent with $\theta_* = 22.4$. ✓
- $V_a \approx 92.47\,\mathrm{m/s}$ **constant.** Trim throttle exactly balances aerodynamic drag. ✓
- β, p, r **at** $\mathcal{O}(10^{-12})$**–**$\mathcal{O}(10^{-13})$**.** Lateral states identically zero; symmetry confirmed. ✓
- q **at** $\mathcal{O}(10^{-11})$ **— zero.** No unbalanced pitching moment. ✓
- $\delta_e \approx -30$ **constant.** Moment balance: $\delta_{e,*} = -C_{m_\alpha}\alpha_*/C_{m_{\delta_e}} = -(-1.2)(0.41)/(-0.9) \approx -0.55\,\mathrm{rad} \approx -31$, consistent with the observed value. ✓
- $\delta_t \approx 0.5$ **constant.** $\approx 50\%$ throttle overcomes drag at trim. ✓
- $\delta_a = \delta_r = 0$**.** Symmetric trim; no lateral control needed. ✓

5.3 Transfer Function Models (`compute_tf_model.m`)

5.3.1 Longitudinal Channel

Pitch angle / elevator.

$$\frac{\theta(s)}{\delta_e(s)} = \frac{a_{\theta_3}}{s^2 + a_{\theta_1}s + a_{\theta_2}}, \qquad a_{\theta_1} = -\frac{\bar{q}_* Sc\, C_{m_q} \cdot c}{2J_y V_{a,*}}, \quad a_{\theta_2} = -\frac{\bar{q}_* Sc\, C_{m_\alpha}}{J_y}, \quad a_{\theta_3} = \frac{\bar{q}_* Sc\, C_{m_{\delta_e}}}{J_y}. \tag{20}$$

Airspeed / throttle, altitude / pitch, airspeed / pitch.

$$\frac{V_a(s)}{\delta_t(s)} = \frac{a_{V_2}}{s + a_{V_1}}, \qquad \frac{h(s)}{\theta(s)} = \frac{V_{a,*}}{s}, \qquad \frac{V_a(s)}{\theta(s)} = \frac{-a_{V_3}}{s + a_{V_1}}, \quad a_{V_3} = g\cos(\theta_* - \alpha_*). \tag{21}$$

5.3.2 Lateral Channel

Roll / aileron, course / roll, sideslip / rudder.

$$\frac{\phi(s)}{\delta_a(s)} = \frac{a_{\phi_2}}{s^2 + a_{\phi_1}s}, \qquad \frac{\chi(s)}{\phi(s)} = \frac{g/V_{a,*}}{s}, \qquad \frac{\beta(s)}{\delta_r(s)} = \frac{a_{\beta_2}}{s + a_{\beta_1}}. \tag{22}$$

5.3.3 Numerical Coefficients

All values below are computed by `compute_tf_model.m` at the trim condition $V_{a,*} = 92.47\,\mathrm{m/s}$, $\theta_* = 0.3906\,\mathrm{rad}$.

Table 5: Transfer function coefficients at trim ($V_{a,*} = 92.47\,\mathrm{m/s}$, $\gamma = 0$, $R = \infty$)

Channel	Parameter	Value	Unit
Longitudinal	a_{θ_1}	0.2947	s^{-1}
	a_{θ_2}	3.3700	s^{-2}
	a_{θ_3}	−2.5275	$\mathrm{rad}^{-1}\,\mathrm{s}^{-2}$
	a_{V_1}	0.0489	s^{-1}
	a_{V_2}	16.0336	$\mathrm{m\,s}^{-2}$
	a_{V_3}	9.8100	$\mathrm{m\,s}^{-2}\,\mathrm{rad}^{-1}$
Lateral	a_{ϕ_1}	1.7305	s^{-1}
	a_{ϕ_2}	−8.0887	$\mathrm{rad}^{-1}\,\mathrm{s}^{-2}$
	a_{β_1}	0.1040	s^{-1}
	a_{β_2}	0.0334	$\mathrm{rad}^{-1}\,\mathrm{s}^{-2}$

Remarks on the coefficients. $a_{\theta_2} = 3.37\,\mathrm{s}^{-2} > 0$ confirms positive pitch stiffness (i.e. static stability of the pitch axis is maintained at trim even though $C_{m_\alpha} < 0$, because the feedback of $k_{p,\theta} a_{\theta_3}$ dominates). $a_{\phi_2} = -8.09$ is negative because the aileron power $C_{\ell_{\delta_a}} < 0$ with the chosen sign convention (positive aileron deflection rolls the aircraft left). $a_{V_1} = 0.049$ is very small, indicating weak aerodynamic drag damping on airspeed — consistent with the X-15's low drag coefficient at this condition.

5.4 State-Space Models (`compute_ss_model.m`)

The full linearised model is obtained via `[A,B,C,D] = linmod('mavsim_trim', x_trim, u_trim)`. Sifting matrices extract decoupled subsystems.

Lateral. States $\{v, p, r, \phi, \psi\}$ selected by $E_1 \in \mathbb{R}^{5\times 12}$; inputs $\{\delta_a, \delta_r\}$ by $E_2 \in \mathbb{R}^{2\times 4}$: $A_{\mathrm{lat}} = E_1 A E_1^\top$, $B_{\mathrm{lat}} = E_1 B E_2^\top$.

Longitudinal. States $\{u, w, q, \theta, p_d\}$ selected by $E_3 \in \mathbb{R}^{5\times 12}$; inputs $\{\delta_e, \delta_t\}$ by $E_4 \in \mathbb{R}^{2\times 4}$: $A_{\mathrm{lon}} = E_3 A E_3^\top$, $B_{\mathrm{lon}} = E_3 B E_4^\top$.

The natural modes (short-period, phugoid, Dutch roll, roll subsidence, spiral) appear as eigenvalues of A_{lon} and A_{lat} respectively and can be extracted in MATLAB via `eig(A_lon)` and `eig(A_lat)`.

6 Section 5: Autopilot Design

6.1 Design Methodology: Successive Loop Closure

All loops are designed using cascaded successive loop closure with bandwidth separation. An inner loop with natural frequency $\omega_{n,\mathrm{inner}}$ is designed first; the outer loop bandwidth is set 5–12× lower so the inner loop appears instantaneous to the outer loop.

6.2 Lateral Autopilot

6.2.1 Roll Angle Hold (Inner Loop)

Plant: $G_\phi(s) = a_{\phi_2}/(s^2 + a_{\phi_1}s)$. PD + integrator (`roll_with_aileron`):

$$\delta_a = k_{p,\phi}(\phi_c - \phi) - k_{d,\phi}p + k_{i,\phi}\int(\phi_c - \phi)\,dt, \tag{23}$$

saturated to $|\delta_a| \le 40$. With $\zeta_\phi = 0.707$, $\delta_{a,\max} = 40$, $e_{\phi,\max} = 15$:

$$k_{p,\phi} = \operatorname{sgn}(a_{\phi_2})\frac{\delta_{a,\max}}{e_{\phi,\max}} = -2.6667, \quad \omega_{n,\phi} = \sqrt{|a_{\phi_2}||k_{p,\phi}|} = \sqrt{8.0887 \times 2.6667} = 4.644\,\text{rad/s}, \tag{24}$$

$$k_{d,\phi} = \frac{2\zeta_\phi\omega_{n,\phi} - a_{\phi_1}}{a_{\phi_2}} = \frac{2(0.707)(4.644) - 1.7305}{-8.0887} = -0.5979, \quad k_{i,\phi} = 0.01. \tag{25}$$

6.2.2 Course Angle Hold (Outer Loop)

Plant: $G_\chi(s) = (g/V_a)/s$. PI controller (`course_with_roll`), saturated to $|\phi_c| \le 40$:

$$\begin{aligned} \omega_{n,\chi} &= \frac{\omega_{n,\phi}}{6} = \frac{4.644}{6} = 0.774\,\text{rad/s}, \\ k_{p,\chi} &= \frac{2\zeta_\chi\omega_{n,\chi}V_{a,*}}{g} = \frac{2(1.0)(0.774)(92.47)}{9.81} = 14.60, \\ k_{i,\chi} &= \frac{\omega_{n,\chi}^2 V_{a,*}}{g} = \frac{(0.774)^2(92.47)}{9.81} = 5.66. \end{aligned} \tag{26}$$

6.2.3 Sideslip Regulation

$$\begin{aligned} k_{p,\theta} &= \operatorname{sgn}(a_{\theta_3})\frac{\delta_{e,\max}}{e_{\theta,\max}} = -2.0000, \\ \omega_{n,\theta} &= \sqrt{a_{\theta_2} + k_{p,\theta}a_{\theta_3}} = \sqrt{3.370 + (-2.0)(-2.5275)} = \sqrt{8.425} = 2.903\,\text{rad/s}, \end{aligned} \tag{27}$$

$$\begin{aligned} k_{d,\theta} &= \frac{2\zeta_\theta\omega_{n,\theta} - a_{\theta_1}}{a_{\theta_3}} = \frac{2(0.707)(2.903) - 0.2947}{-2.5275} = -1.5072, \\ K_{\theta,\text{DC}} &= \frac{a_{\theta_3}k_{p,\theta}}{a_{\theta_2} + a_{\theta_3}k_{p,\theta}} = 0.6000. \end{aligned} \tag{28}$$

$$\begin{aligned} \omega_{n,h} &= \frac{\omega_{n,\theta}}{12} = \frac{2.903}{12} = 0.242\,\text{rad/s}, \\ k_{p,h} &= \frac{2\zeta_h\omega_{n,h}}{|K_{\theta,\text{DC}}|V_{a,*}} = \frac{2(1.0)(0.242)}{0.6 \times 92.47} = 0.008719, \\ k_{i,h} &= \frac{\omega_{n,h}^2}{|K_{\theta,\text{DC}}|V_{a,*}} = 0.001055. \end{aligned} \tag{29}$$

6.2.4 Airspeed Hold via Throttle

PI controller (`airspeed_with_throttle`), $\delta_t \in [0, 1]$, $\omega_{n,V_a} = 0.3\,\text{rad/s}$:

$$k_{p,V_a} = \frac{2\zeta_{V_a}\omega_{n,V_a} - a_{V_1}}{a_{V_2}} = \frac{2(0.707)(0.3) - 0.0489}{16.0336} = 0.0234, \quad k_{i,V_a} = \frac{\omega_{n,V_a}^2}{a_{V_2}} = \frac{0.09}{16.0336} = 0.005613. \tag{30}$$

6.2.5 Airspeed Hold via Commanded Pitch

Exploiting $T_{V_a/\theta} = -a_{V_3}/(s + a_{V_1})$:

$$\theta_c^{V_a} = k_{p,V_a^2}(V_{a,c} - V_a) + k_{i,V_a^2}\int (V_{a,c} - V_a)\,dt, \quad k_{p,V_a^2} = 0.0548, \quad k_{i,V_a^2} = 0.017527. \tag{31}$$

6.3 Autopilot Gain Summary

Table 6: Autopilot gains from `compute_autopilot_gains.m` at $V_{a,*} = 92.47\,\mathrm{m/s}$

Loop	Gain	Value	Notes
Roll (inner)	$k_{p,\phi}$	-2.6667	$\delta_{a,\max} = 40,\ e_{\phi,\max} = 15$
	$k_{d,\phi}$	-0.5979	$\omega_{n,\phi} = 4.644\,\mathrm{rad/s}, \zeta = 0.707$
	$k_{i,\phi}$	0.01	
Course (outer)	$k_{p,\chi}$	14.60	$\omega_{n,\chi} = 0.774\,\mathrm{rad/s}$
	$k_{i,\chi}$	5.66	
Pitch (inner)	$k_{p,\theta}$	-2.0000	$\delta_{e,\max} = 40,\ e_{\theta,\max} = 20$
	$k_{d,\theta}$	-1.5072	$\omega_{n,\theta} = 2.903\,\mathrm{rad/s}, \zeta = 0.707$
	$K_{\theta,\mathrm{DC}}$	0.6000	
Altitude (outer)	$k_{p,h}$	0.008719	$\omega_{n,h} = 0.242\,\mathrm{rad/s}$
	$k_{i,h}$	0.001055	
Va / throttle	k_{p,V_a}	0.0234	$\omega_{n,V_a} = 0.3\,\mathrm{rad/s}$
	k_{i,V_a}	0.005613	
Va / pitch	k_{p,V_a^2}	0.0548	
	k_{i,V_a^2}	0.017527	

6.4 Expected Simulation Results

Step-response plots for the autopilot are to be added once the Simulink autopilot simulation is complete. The expected qualitative behaviour for each required plot is described below.

Lateral autopilot.

- **Roll ϕ without disturbance.** Step command ϕ_c should be tracked with $\approx 5\%$ overshoot (from $\zeta_\phi = 0.707$) and settling time $\approx 4/(\zeta_\phi \omega_{n,\phi}) \approx 1.22\,\mathrm{s}$.
- **Roll ϕ with step disturbance (magnitude 0.2).** The integral term $k_{i,\phi} = 0.01$ eliminates steady-state error; recovery should be visible within a few seconds.
- **Course angle χ.** The outer loop is 6× slower ($\omega_{n,\chi} = 0.774\,\mathrm{rad/s}$); expect a longer settling time of $\approx 7\,\mathrm{s}$ with an intermediate roll transient.
- **Sideslip β.** With $\delta_r = 0$, β remains small due to passive directional stability ($C_{n_\beta} = 0.5$).
- **Aileron δ_a.** Saturation at ±40 may be briefly active at the roll step; the control returns to linear range within a fraction of the settling time.

Longitudinal autopilot.

- **Pitch θ.** PD loop ($\omega_{n,\theta} = 2.903\,\mathrm{rad/s}$) tracks rapidly with $\approx 5\%$ overshoot, settling in $\approx 2\,\mathrm{s}$.

- **Altitude h.** Outer loop ($\omega_{n,h} = 0.242\,\mathrm{rad/s}$) is much slower; settling in $\approx 25\,\mathrm{s}$. The altitude zone of 10 m limits pitch commands and prevents aggressive pitch manoeuvres.
- **Airspeed V_a via throttle.** At $\omega_{n,V_a} = 0.3\,\mathrm{rad/s}$, the settling time is $\approx 13\,\mathrm{s}$.
- **Airspeed V_a via pitch.** Faster than throttle channel but creates coupling with altitude.
- **Elevator δ_e and throttle δ_t.** Elevator saturation at ± 40 may be briefly active at pitch step transitions; throttle remains in $[0, 1]$.

7 Discussion and Conclusions

This project implemented a complete 6-DOF simulation and autopilot pipeline for the North American X-15, verifying each stage systematically.

1. **Animation (Figs. 1, 3).** The vertex-face model renders correctly. Rotation-before-translation is confirmed essential for correct body-frame orientation.

2. **Equations of motion (Fig. 2).** A pure body-x force produces parabolic p_n growth with zero cross-axis contamination, verifying Eqs. (4)–(9).

3. **Forces and moments (Fig. 4).** At zero control inputs the aircraft pitches down to $\theta \approx -40$ due to the unbalanced pitching moment. Lateral states remain decoupled at zero throughout, confirming lateral–longitudinal separation at $\beta = 0$.

4. **Trim and linearisation (Fig. 5).** Trim converges with residual 4.96×10^{-11} at $V_{a,*} = 92.47\,\mathrm{m/s}$, $\theta_* = 22.4$. The large trim angle is physically explained by the X-15's high wing loading. Transfer function coefficients (table 5) and autopilot gains (table 6) are all computed from this trim point.

5. **Autopilot (Sec. 6).** Successive loop closure with bandwidth ratios $\omega_{n,\phi} : \omega_{n,\chi} = 6 : 1$ and $\omega_{n,\theta} : \omega_{n,h} = 12 : 1$ is designed. All gains are numerically determined. Step-response simulation plots are to be added.

Assumptions.

- Zero-offset aerodynamic coefficients assumed (symmetric aircraft).
- Rocket thrust proportional to throttle; fuel-mass depletion neglected.
- Standard sea-level atmosphere ($\rho = 1.225\,\mathrm{kg/m^3}$).
- Kinematic singularity at $\theta = \pm 90$ not encountered.

A Key MATLAB Code Extracts

A.1 Rotational Dynamics (`mav_dynamics.m`)

```
p_dot = P.Gamma1*p*q - P.Gamma2*q*r + P.Gamma3*ell + P.Gamma4*n;
q_dot = P.Gamma5*p*r - P.Gamma6*(p^2 - r^2) + m/P.Jy;
r_dot = P.Gamma7*p*q - P.Gamma1*q*r + P.Gamma4*ell + P.Gamma8*n;
```

Listing 1: Eqs. (3.17): rotational rate dynamics with J_{xz} coupling

A.2 Lift/Drag to Body Frame (`forces_moments.m`)

```
CL = MAV.C_L_0 + MAV.C_L_alpha*alpha ...
   + (MAV.C_L_q*MAV.c*q)/(2*Va) + MAV.C_L_delta_e*delta_e;
CD = MAV.C_D_0 + MAV.C_D_alpha*alpha ...
   + (MAV.C_D_q*MAV.c*q)/(2*Va) + MAV.C_D_delta_e*delta_e;
x_aero = q_bar*MAV.S_wing*(-CD*cos(alpha) + CL*sin(alpha));
z_aero = q_bar*MAV.S_wing*(-CD*sin(alpha) - CL*cos(alpha));
```

Listing 2: Wind-frame force coefficients resolved into body frame

A.3 Roll and Pitch Gain Design (`compute_autopilot_gains.m`)

```
% Roll
AP.roll_kp = sign(a_phi2) * delta_a_max / e_phi_max;      % = -2.6667
wn_phi     = sqrt(abs(a_phi2) * abs(AP.roll_kp));          % = 4.644 rad/s
AP.roll_kd = (2*zeta_phi*wn_phi - a_phi1) / a_phi2;       % = -0.5979
AP.roll_ki = 0.01;
% Pitch
AP.pitch_kp  = sign(a_theta3) * delta_e_max / e_theta_max; % = -2.0000
wn_theta     = sqrt(a_theta2 + AP.pitch_kp * a_theta3);    % = 2.903 rad/s
AP.pitch_kd  = (2*zeta_theta*wn_theta - a_theta1) / a_theta3; % = -1.5072
AP.K_theta_DC = a_theta3*AP.pitch_kp / (a_theta2 + a_theta3*AP.pitch_kp); %
     = 0.6000
```

Listing 3: Bandwidth-placement gains for roll and pitch loops

References

[1] R. W. Beard and T. W. McLain, *Small Unmanned Aircraft: Theory and Practice*. Princeton University Press, 2012.

[2] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft Control and Simulation*, 3rd ed. Wiley, 2015.